

30 CHAPTERWISE IMPORTANT QUESTIONS

Mathematics Standard (041) | CBSE 2026-27

Pair of Linear Equations in Two Variables

CBSE Mathematics 2026-27 | Board-examinable

20 numerical / problem-solving questions and 10 theory, proof or method questions. Workings and answers are written for Class X revision.

Question 1

Solve $2x + 3y = 13$ and $3x - 1y = 3$.

Answer:

Eliminating one variable gives $x = 2$ and $y = 3$. Substitution verifies both equations.

Question 2

Solve $4x + 1y = 14$ and $1x - 2y = -1$.

Answer:

Eliminating one variable gives $x = 3$ and $y = 2$. Substitution verifies both equations.

Question 3

Solve $3x + 2y = 11$ and $5x - 1y = 1$.

Answer:

Eliminating one variable gives $x = 1$ and $y = 4$. Substitution verifies both equations.

Question 4

Solve $5x - 2y = 8$ and $2x + 3y = 7$.

Answer:

Eliminating one variable gives $x = 2$ and $y = 1$. Substitution verifies both equations.

Question 5

Solve $1x + 4y = 14$ and $3x + 2y = 12$.

Answer:

Eliminating one variable gives $x = 2$ and $y = 3$. Substitution verifies both equations.

Question 6

Solve $6x + 1y = 11$ and $2x - 3y = -13$.

Answer:

Eliminating one variable gives $x = 1$ and $y = 5$. Substitution verifies both equations.

Question 7

Solve $2x + 5y = 11$ and $4x - 1y = 11$.

Answer:

Eliminating one variable gives $x = 3$ and $y = 1$. Substitution verifies both equations.

Question 8

Solve $7x - 2y = 6$ and $1x + 3y = 14$.

Answer:

Eliminating one variable gives $x = 2$ and $y = 4$. Substitution verifies both equations.

Question 9

Solve $3x - 4y = 2$ and $5x + 2y = 12$.

Answer:

Eliminating one variable gives $x = 2$ and $y = 1$. Substitution verifies both equations.

Pair of Linear Equations in Two Variables - Continued

Question 10

Solve $4x + 3y = 10$ and $2x + 5y = 12$.

Answer:

Eliminating one variable gives $x = 1$ and $y = 2$. Substitution verifies both equations.

Question 11

Classify the pair $2x + 4y = 8$ and $1x + 2y = 4$.

Answer:

The coefficient ratios are equal. The constant ratio is also equal, so the pair has infinitely many solutions.

Question 12

Classify the pair $3x + 6y = 5$ and $1x + 2y = 6$.

Answer:

The coefficient ratios are equal. The constant ratio is different, so the pair has no solution.

Question 13

Classify the pair $5x + 10y = 15$ and $1x + 2y = 3$.

Answer:

The coefficient ratios are equal. The constant ratio is also equal, so the pair has infinitely many solutions.

Question 14

Classify the pair $4x + 8y = 7$ and $1x + 2y = 8$.

Answer:

The coefficient ratios are equal. The constant ratio is different, so the pair has no solution.

Question 15

Classify the pair $7x + 14y = 21$ and $1x + 2y = 3$.

Answer:

The coefficient ratios are equal. The constant ratio is also equal, so the pair has infinitely many solutions.

Question 16

Adult tickets cost Rs 50 and child tickets cost Rs 30. 10 tickets cost Rs 420. Find each type.

Answer:

Let adult tickets = x . Then $x + y = 10$ and $50x + 30y = 420$. Solving gives $x = 6$ adults and $y = 4$ children.

Question 17

Adult tickets cost Rs 40 and child tickets cost Rs 25. 12 tickets cost Rs 405. Find each type.

Answer:

Let adult tickets = x . Then $x + y = 12$ and $40x + 25y = 405$. Solving gives $x = 7$ adults and $y = 5$ children.

Question 18

Adult tickets cost Rs 60 and child tickets cost Rs 35. 15 tickets cost Rs 750. Find each type.

Answer:

Let adult tickets = x . Then $x + y = 15$ and $60x + 35y = 750$. Solving gives $x = 9$ adults and $y = 6$ children.

Pair of Linear Equations in Two Variables - Continued

Question 19

Adult tickets cost Rs 80 and child tickets cost Rs 50. 20 tickets cost Rs 1330. Find each type.

Answer:

Let adult tickets = x . Then $x + y = 20$ and $80x + 50y = 1330$. Solving gives $x = 11$ adults and $y = 9$ children.

Question 20

Adult tickets cost Rs 70 and child tickets cost Rs 40. 18 tickets cost Rs 960. Find each type.

Answer:

Let adult tickets = x . Then $x + y = 18$ and $70x + 40y = 960$. Solving gives $x = 8$ adults and $y = 10$ children.

Question 21

State the main formula, rule or theorem used in Pair of Linear Equations in Two Variables.

Answer:

For $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$, compare a_1/a_2 , b_1/b_2 and c_1/c_2 .

Question 22

What do the important symbols or quantities in Pair of Linear Equations in Two Variables represent?

Answer:

A solution is an ordered pair satisfying both equations.

Question 23

Write the correct method for solving a Pair of Linear Equations in Two Variables numerical.

Answer:

Use elimination, substitution, cross multiplication or a graph, then verify in both equations.

Question 24

Which unit, sign convention or condition must be checked in a Pair of Linear Equations in Two Variables answer?

Answer:

Write the given data first, use the stated convention, and include units wherever the final quantity has a unit.

Question 25

Why must the final answer in Pair of Linear Equations in Two Variables be checked against the given condition?

Answer:

Checking catches wrong signs, arithmetic errors and values that are impossible in the original situation.

Pair of Linear Equations in Two Variables - Continued

Question 26

What is one common error in Pair of Linear Equations in Two Variables questions?

Answer:

Students often substitute a value in the wrong formula or skip a condition. Write the formula before substituting.

Question 27

How should full working be shown in a Pair of Linear Equations in Two Variables numerical?

Answer:

Show the formula or theorem, substitution, simplification, calculation and final answer in that order.

Question 28

When is a diagram, table or graph useful in Pair of Linear Equations in Two Variables?

Answer:

Use it when it makes the given quantities and their relationship clear before calculation.

Question 29

How can a student decide which formula to use in Pair of Linear Equations in Two Variables?

Answer:

Identify the quantity asked, list the known quantities, then select the formula or theorem connecting them.

Question 30

What makes a high-scoring answer in Pair of Linear Equations in Two Variables?

Answer:

Correct reasoning, complete steps, an appropriate formula, a clearly stated final result and a quick verification.

Sources: CBSE Mathematics curriculum 2026-27; CBSE Class X sample papers and marking schemes 2026-27; NCERT textbooks.